DIMENSION AND LOCAL RING DIMENSION ARE INVARIANT UNDER ETALE MORPHISMS OF DECENT SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original definition 04N5: sources/stacks-chow/tags.txt, source lines 6066–6066. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 04N6: sources/stacks-chow/tags.txt, source lines 6067–6067. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be an étale morphism of algebraic spaces over S. Let $x \in X$. Then (1) $\dim_x(X) = \dim_{f(x)}(Y)$ and (2) the dimension of the local ring of X at x equals the dimension of the local ring of Y at $f(x)$. If f is surjective, then (3) $\dim(X) = \dim(Y)$.*

Proof. Choose a scheme U and a point $u \in U$ and an étale morphism $U \to X$ which maps u to x. Then the composition $U \to Y$ is also étale and maps u to $f(x)$. Thus the statements (1) and (2) follow as the relevant integers are defined in terms of the behaviour of the scheme U at u. See Definition 04N5 for (1). Part (3) is an immediate consequence of (1), see Definition 04N6. □

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.